Answers

(a) An atom of ^{222}Rn has 86 electrons outside the nucleus. Inside the nucleus are 86 protons and 222 − 86 = 136 neutrons. An atom of ^{220}Rn has the same number of electrons and protons, but only 134 neutrons in its nucleus.

(b) Four alpha particles are produced to make ^{222}Rn from ^{238}U; three are produced to make ^{220}Rn from ^{232}Th.

(c) If $^{238}_{92}\text{U}$ were to lose four alpha particles, $^{222}_{84}\text{Po}$ would result instead of $^{222}_{84}\text{Rn}$. Two $^{0}_{-1}e^{-}$ beta particles must be ejected from the nucleus along the way to raise the atomic number to $Z = 86$. The same is true of the production of ^{220}Rn from ^{232}Th.

(d) $^{222}_{86}\text{Rn} \longrightarrow {}^{218}_{84}\text{Po} + {}^{4}_{2}\text{He}$; $\Delta m = -0.00617$ u < 0; allowed
$^{220}_{86}\text{Rn} \longrightarrow {}^{216}_{84}\text{Po} + {}^{4}_{2}\text{He}$; $\Delta m = -0.0069$ u < 0; allowed

(e) $\Delta E = -6.4 \text{ MeV} = -1.03 \times 10^{-12}$ J

(f) $A = 1.14 \times 10^{8}\ \text{s}^{-1}$

(g) $A = 9.0 \times 10^{6}\ \text{s}^{-1}$

(h) Greater

PROBLEMS

Answers to problems whose numbers are boldface appear in Appendix G. Problems that are more challenging are indicated with asterisks.

Nuclear Structure and Nuclear Decay Processes

1. The nuclide $^{8}_{5}\text{B}$ decays by positron emission to $^{8}_{4}\text{Be}$. What is the energy released (in MeV)?

2. The nuclide $^{10}_{4}\text{Be}$ undergoes spontaneous radioactive decay to $^{10}_{5}\text{B}$ with emission of a beta particle. Calculate the maximum kinetic energy of the emitted beta particle.

3. Write balanced equations that represent the following nuclear reactions.
(a) Beta emission by $^{39}_{17}\text{Cl}$
(b) Positron emission by $^{22}_{11}\text{Na}$
(c) Alpha emission by $^{224}_{88}\text{Ra}$
(d) Electron capture by $^{82}_{38}\text{Sr}$

4. Write balanced equations that represent the following nuclear reactions.
(a) Alpha emission by $^{155}_{70}\text{Yb}$
(b) Positron emission by $^{26}_{14}\text{Si}$
(c) Electron capture by $^{65}_{30}\text{Zn}$
(d) Beta emission by $^{100}_{41}\text{Nb}$

5. The stable isotopes of neon are ^{20}Ne, ^{21}Ne, and ^{22}Ne. Predict the nuclides formed when ^{19}Ne and ^{23}Ne decay.

6. The two stable isotopes of carbon are ^{12}C and ^{13}C. Predict the nuclides formed when ^{11}C and ^{14}C decay. Is alpha emission by ^{14}C possible?

7. The free neutron is an unstable particle that decays into a proton. What other particle is formed in neutron decay, and what is the maximum kinetic energy (in MeV) that it can possess?

8. The radionuclide $^{210}_{84}\text{Po}$ decays by alpha emission to a daughter nuclide. The atomic mass of $^{210}_{84}\text{Po}$ is 209.9829 u, and that of its daughter is 205.9745 u.
(a) Identify the daughter, and write the nuclear equation for the radioactive decay process.
(b) Calculate the total energy released per disintegration (in MeV).
(c) Calculate the kinetic energy of the emitted alpha particle.

9. The natural abundance of ^{30}Si is 3.1%. Upon irradiation with neutrons, this isotope is converted to ^{31}Si, which decays to the stable isotope ^{31}P. This provides a way of introducing trace amounts of phosphorus into silicon in a much more uniform fashion than is possible by ordinary mixing of silicon and phosphorus and gives semiconductor devices the capability of handling much higher levels of power. Write balanced nuclear equations for the two steps in the preparation of ^{31}P from ^{30}Si.

10. The most convenient way to prepare the element polonium is to expose bismuth (which is 100% ^{209}Bi) to neutrons. Write balanced nuclear equations for the two steps in the preparation of polonium.

11. One convenient source of neutrons is the reaction of an alpha particle from an emitter such as polonium (^{210}Po) with an atom of beryllium (^{9}Be). Write nuclear equations for the reactions that occur.

12. Three atoms of element 111 were produced in 1994 by bombarding ^{209}Bi with ^{64}Ni.
(a) Write a balanced equation for this nuclear reaction. What other species is produced?
(b) Write a balanced equation for the alpha decay process of this nuclide of element 111.

Mass–Energy Relationships

13. Complete and balance the following equations for nuclear reactions that are thought to take place in stars:
(a) $2\ {}^{12}_{6}\text{C} \longrightarrow ? + {}^{1}_{0}n$
(b) $? + {}^{1}_{1}\text{H} \longrightarrow {}^{12}_{6}\text{C} + {}^{4}_{2}\text{He}$
(c) $2\ {}^{3}_{2}\text{He} \longrightarrow ? + 2\ {}^{1}_{1}\text{H}$

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14. Complete and balance the following equations for nuclear reactions that are used in particle accelerators to make elements beyond uranium:
(a) ${}^{4}_{2}He + {}^{253}_{99}Es \longrightarrow ? + 2{}^{1}_{0}n$
(b) ${}^{249}_{98}Cf + ? \longrightarrow {}^{257}_{103}Lr + 2{}^{1}_{0}n$
(c) ${}^{238}_{92}U + {}^{12}_{6}C \longrightarrow {}^{244}_{98}Cf + ?$

15. Calculate the total binding energy, in both kJ per mole and MeV per atom, and the binding energy per nucleon of the following nuclides, using the data from Table 19.1.
(a) ${}^{40}_{20}Ca$ (b) ${}^{87}_{37}Rb$ (c) ${}^{238}_{92}U$

16. Calculate the total binding energy, in both kilojoules per mole and MeV per atom, and the binding energy per nucleon of the following nuclides, using the data from Table 19.1.
(a) ${}^{10}_{4}Be$ (b) ${}^{35}_{17}Cl$ (c) ${}^{49}_{22}Ti$

17. Use the data from Table 19.1 to predict which is more stable: four protons, four neutrons, and four electrons organized as two ${}^{4}He$ atoms or as one ${}^{8}Be$ atom. What is the mass difference?

18. Use the data from Table 19.1 to predict which is more stable: 16 protons, 16 neutrons, and 16 electrons organized as two ${}^{16}O$ atoms or as one ${}^{32}S$ atom. What is the mass difference?

Kinetics of Radioactive Decay

19. How many radioactive disintegrations occur per minute in a 0.0010-g sample of ${}^{209}Po$ that has been freshly separated from its decay products? The half-life of ${}^{209}Po$ is 103 years.

20. How many alpha particles are emitted per minute by a 0.0010-g sample of ${}^{238}U$ that has been freshly separated from its decay products? Assume that each decay emits one alpha particle. The half-life of ${}^{238}U$ is 4.47×10^{9} years.

21. The nuclide ${}^{19}O$, prepared by neutron irradiation of ${}^{19}F$, has a half-life of 29 s.
(a) How many ${}^{19}O$ atoms are in a freshly prepared sample if its decay rate is $2.5 \times 10^{4}\ s^{-1}$?
(b) After 2.00 min, how many ${}^{19}O$ atoms remain?

22. The nuclide ${}^{35}S$ decays by beta emission with a half-life of 87.1 days.
(a) How many grams of ${}^{35}S$ are in a sample that has a decay rate from that nuclide of $3.70 \times 10^{2}\ s^{-1}$?
(b) After 365 days, how many grams of ${}^{35}S$ remain?

23. Astatine is the rarest naturally occurring element, with ${}^{219}At$ appearing as the product of a very minor side branch in the decay of ${}^{235}U$ (itself not a very abundant isotope). It is estimated that the mass of all the naturally occurring ${}^{219}At$ in the upper kilometer of the earth's surface has a steady-state value of only 44 mg. Calculate the total activity (in disintegrations per second) caused by all the naturally occurring astatine in this part of the earth. The half-life of ${}^{219}At$ is 54 s, and its atomic mass is 219.01 u.

24. Technetium has not been found in nature. It can be obtained readily as a product of uranium fission in nuclear power plants, however, and is now produced in quantities of many kilograms per year. One medical use relies on the tendency of ${}^{99m}Tc$ (an excited nuclear state of ${}^{99}Tc$) to concentrate in abnormal heart tissue. Calculate the total activity (in disintegrations per second) caused by the decay of 1.0 μg of ${}^{99m}Tc$, which has a half-life of 6.0 hours.

25. The specific activity of ${}^{14}C$ in the biosphere is 0.255 Bq g^{-1}. What is the age of a piece of papyrus from an Egyptian tomb if its beta counting rate is 0.153 Bq g^{-1}? The half-life of ${}^{14}C$ is 5730 years.

26. The specific activity of an article found in the Lascaux Caves in France is 0.0375 Bq g^{-1}. Calculate the age of the article.

27. Over geological time, an atom of ${}^{238}U$ decays to a stable ${}^{206}Pb$ atom in a series of eight alpha emissions, each of which leads to the formation of one helium atom. A geochemist analyzes a rock and finds that it contains $9.0 \times 10^{-5}\ cm^{3}$ of helium (at 0°C and atmospheric pressure) per gram and 2.0×10^{-7} g of ${}^{238}U$ per gram. Estimate the age of the mineral, given that $t_{1/2}$ of ${}^{238}U$ is 4.47×10^{9} years.

28. The isotope ${}^{232}Th$ decays to ${}^{208}Pb$ by the emission of six alpha particles, with a half-life of 1.39×10^{10} years. Analysis of 1.00 kg of ocean sediment shows it to contain 7.4 mg of ${}^{232}Th$ and $4.9 \times 10^{-3}\ cm^{3}$ of gaseous helium at 0°C and atmospheric pressure. Estimate the age of the sediment, assuming no loss or gain of thorium or helium from the sediment since its formation and assuming that the helium arose entirely from the decay of thorium.

29. The half-lives of ${}^{235}U$ and ${}^{238}U$ are 7.04×10^{8} years and 4.47×10^{9} years, respectively, and the present abundance ratio is ${}^{238}U/{}^{235}U = 137.7$. It is thought that their abundance ratio was 1 at some time *before* our earth and solar system were formed about 4.5×10^{9} years ago. Estimate how long ago the supernova occurred that supposedly produced all the uranium isotopes in equal abundance, including the two longest lived isotopes, ${}^{238}U$ and ${}^{235}U$.

30. Using the result of problem 29 and the accepted age of the earth, 4.5×10^{9} yr, calculate the ${}^{238}U/{}^{235}U$ ratio at the time the earth was formed.

Radiation in Biology and Medicine

31. Write balanced equations for the decays of ${}^{11}C$ and ${}^{15}O$, both of which are used in positron emission tomography to scan the uptake of glucose in the body.

32. Write balanced equations for the decays of ${}^{13}N$ and ${}^{18}F$, two other radioisotopes that are used in positron emission tomography. What is the ultimate fate of the positrons?

33. The positrons emitted by ${}^{11}C$ have a maximum kinetic energy of 0.99 MeV, and those emitted by ${}^{15}O$ have a maximum kinetic energy of 1.72 MeV. Calculate the ratio of the number of millisieverts of radiation exposure caused by ingesting a given fixed chemical amount (equal numbers of atoms) of each of these radioisotopes.

34. Compare the relative health risks of contact with a given amount of ${}^{226}Ra$, which has a half-life of 1622 years and emits 4.78-MeV alpha particles, with contact with the same chemical amount of ${}^{14}C$, which has a half-life of 5730 years and emits beta particles with energies of up to 0.155 MeV.

35. The nuclide ${}^{131}I$ undergoes beta decay with a half-life of 8.041 days. Large quantities of this nuclide were released into the environment in the Chernobyl accident. A victim of radiation poisoning has absorbed 5.0×10^{-6} g (5.0 μg) of ${}^{131}I$.

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(a) Compute the activity, in becquerels, of the ^{131}I in this person, taking the atomic mass of the nuclide to equal 131 g mol^{-1}.
(b) Compute the radiation absorbed dose, in milligrays, caused by this nuclide during the first *second* after its ingestion. Assume that beta particles emitted by ^{131}I have an average kinetic energy of 0.40 MeV, that all of this energy is deposited within the victim's body, and that the victim weighs 60 kg.
(c) Is this dose likely to be lethal? Remember that the activity of the ^{131}I diminishes as it decays.

36. The nuclide ^{239}Pu undergoes alpha decay with a half-life of 2.411×10^4 years. An atomic energy worker breathes in 5.0×10^{-6} g (5.0 μg) of ^{239}Pu, which lodges permanently in a lung.
(a) Compute the activity, in becquerels, of the ^{239}Pu ingested, taking the atomic mass of the nuclide to be 239 g mol^{-1}.
(b) Determine the radiation absorbed dose, in milligrays, during the first *year* after its ingestion. Assume that alpha particles emitted by ^{239}Pu have an average kinetic energy of 5.24 MeV, that all of this energy is deposited within the worker's body, and that the worker weighs 60 kg.
(c) Is this dose likely to be lethal?

Nuclear Fission

37. Strontium-90 is one of the most hazardous products of atomic weapons testing because of its long half-life ($t_{1/2}$ = 28.1 years) and its tendency to accumulate in bone.
(a) Write nuclear equations for the decay of ^{90}Sr via the successive emission of two beta particles.
(b) The atomic mass of ^{90}Sr is 89.9073 u and that of ^{90}Zr is 89.9043 u. Calculate the energy released per ^{90}Sr atom, in MeV, in decaying to ^{90}Zr.
(c) What will be the initial activity of 1.00 g of ^{90}Sr released into the environment, in disintegrations per second?
(d) What activity will the material from part (c) show after 100 years?

38. Plutonium-239 is the fissionable isotope produced in breeder reactors; it is also produced in ordinary nuclear plants and in weapons tests. It is an extremely poisonous substance with a half-life of 24,100 years.
(a) Write an equation for the decay of ^{239}Pu via alpha emission.
(b) The atomic mass of ^{239}Pu is 239.05216 u and that of ^{235}U is 235.04393 u. Calculate the energy released per ^{239}Pu atom, in MeV, in decaying via alpha emission.
(c) What will be the initial activity, in disintegrations per second, of 1.00 g of ^{239}Pu buried in a disposal site for radioactive wastes?
(d) What activity will the material from part (c) show after 100,000 years?

39. The three naturally occurring isotopes of uranium are ^{234}U (half-life 2.5×10^5 years), ^{235}U (half-life 7.0×10^8 years), and ^{238}U (half-life 4.5×10^9 years). As time passes, will the average atomic mass of the uranium in a sample taken from nature increase, decrease, or remain constant?

40. Natural lithium consists of 7.42% ^6Li and 92.58% ^7Li. Much of the tritium (^3_1H) used in experiments with fusion reactions is made by the capture of neutrons by ^6Li atoms.
(a) Write a balanced nuclear equation for the process. What is the other particle produced?
(b) After ^6Li is removed from natural lithium, the remainder is sold for other uses. Is the molar mass of the leftover lithium greater or smaller than that of natural lithium?

41. Calculate the amount of energy released, in kilojoules per *gram* of uranium, in the fission reaction

$$^{235}_{92}U + ^1_0n \longrightarrow ^{94}_{36}Kr + ^{130}_{56}Ba + 3\ ^1_0n$$

Use the atomic masses in Table 19.1. The atomic mass of ^{94}Kr is 93.919 u and that of ^{139}Ba is 138.909 u.

Nuclear Fusion and Nucleosynthesis

42. Calculate the amount of energy released, in kilojoules per *gram* of deuterium (^2H), for the fusion reaction

$$^2_1H + ^2_1H \longrightarrow + ^4_2He$$

Use the atomic masses in Table 19.1. Compare your answer with that from the preceding problem.

ADDITIONAL PROBLEMS

43. When an electron and a positron meet, they are replaced by two gamma rays, called the "annihilation radiation." Calculate the energies of these radiations, assuming that the kinetic energies of the incoming particles are 0.

44. The nuclide $^{231}_{92}U$ converts spontaneously to $^{231}_{91}Pa$.
(a) Write two balanced nuclear equations for this conversion, one if it proceeds by electron capture and the other if it proceeds by positron emission.
(b) Using the nuclidic masses in Table 19.1, calculate the change in mass for each process. Explain why electron capture can occur spontaneously in this case but positron emission cannot.

45. The radioactive nuclide $^{64}_{29}Cu$ decays by beta emission to $^{64}_{30}Zn$ or by positron emission to $^{64}_{28}Ni$. The maximum kinetic energy of the beta particles is 0.58 MeV, and that of the positrons is 0.65 MeV. The mass of the neutral $^{64}_{29}Cu$ atom is 63.92976 u.
(a) Calculate the mass, in atomic mass units, of the neutral $^{64}_{30}Zn$ atom.
(b) Calculate the mass, in atomic mass units, of the neutral $^{64}_{28}Ni$ atom.

46. A puzzling observation that led to the discovery of isotopes was the fact that lead obtained from uranium-containing ores had an atomic mass lower by two full atomic mass units than lead obtained from thorium-containing ores. Explain this result, using the fact that decay of radioactive uranium and thorium to stable lead occurs via alpha and beta emission.

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47. By 1913, the elements radium, actinium, thorium, and uranium had all been discovered, but element 91, between thorium and uranium in the periodic table, was not yet known. The approach used by Meitner and Hahn was to look for the parent that decays to form actinium. Alpha and beta emission are the most important decay pathways among the heavy radioactive elements. What elements would decay to actinium by each of these two pathways? If radium salts show no sign of actinium, what does this suggest about the parent of actinium? What is the origin of the name of element 91, discovered by Meitner and Hahn in 1918?

48. Working in Rutherford's laboratory in 1932, Cockcroft and Walton bombarded a lithium target with 700-keV protons and found that the following reaction occurred:

$$^{7}_{3}\text{Li} + ^{1}_{1}\text{H} \longrightarrow +\,2\ ^{4}_{2}\text{He}$$

Each of the alpha particles was found to have a kinetic energy of 8.5 MeV. This research provided the first experimental test of Einstein's $\Delta E = c^2\Delta m$ relationship. Discuss. Using the atomic masses from Table 19.1, calculate the value of c needed to account for this result.

49. (a) Calculate the binding energy per nucleon in $^{30}_{15}\text{P}$.
(b) The radioactive decay of the $^{30}_{15}\text{P}$ occurs through positron emission. Calculate the maximum kinetic energy carried off by the positron.
(c) The half-life for this decay is 150 s. Calculate the rate constant k and the fraction remaining after 450 s.

50. Selenium-82 undergoes *double* beta decay:

$$^{82}_{34}\text{Se} \longrightarrow ^{82}_{36}\text{Kr} + 2\ ^{0}_{-1}e^{-} + 2\tilde{\nu}$$

This low-probability process occurs with a half-life of 3.5×10^{27} s, one of the longest half-lives ever measured. Estimate the activity in an 82.0-g (1.00 mol) sample of this isotope. How many ^{82}Se nuclei decay in a day?

51. Gallium citrate, which contains the radioactive nuclide ^{67}Ga, is used in medicine as a tumor-seeking agent. Gallium-67 decays with a half-life of 77.9 hours. How much time is required for it to decay to 5.0% of its initial activity?

52. The nuclide ^{241}Am is used in smoke detectors. As it decays (with a half-life of 458 years), the emitted alpha particles ionize the air. When combustion products enter the detector, the number of ions changes and with it the conductivity of the air, setting off an alarm. If the activity of ^{241}Am in the detector is 3×10^4 Bq, calculate the mass of ^{241}Am present.

53. The half-life of ^{14}C is $t_{1/2} = 5730$ years, and 1.00 g of modern wood charcoal has an activity of 0.255 Bq.
(a) Calculate the number of ^{14}C atoms per gram of carbon in modern wood charcoal.
(b) Calculate the fraction of carbon atoms in the biosphere that are ^{14}C.

54. Carbon-14 is produced in the upper atmosphere by the reaction

$$^{14}_{7}\text{N} + ^{1}_{0}n \longrightarrow ^{14}_{6}\text{C} + ^{1}_{1}\text{H}$$

where the neutrons come from nuclear processes induced by cosmic rays. It is estimated that the steady-state ^{14}C activity in the biosphere is 1.1×10^{19} Bq.
(a) Estimate the total mass of carbon in the biosphere, using the data in problem 53.
(b) The earth's crust has an average carbon content of 250 parts per million by mass, and the total crustal mass is 2.9×10^{25} g. Estimate the fraction of the carbon in the earth's crust that is part of the biosphere. Speculate on the whereabouts of the rest of the carbon in the earth's crust.

55. Analysis of a rock sample shows that it contains 0.42 mg of ^{40}Ar for every 1.00 mg of ^{40}K. Assuming that all the argon resulted from decay of the potassium and that neither element has left or entered the rock since its formation, estimate the age of the rock. (*Hint:* Use data from Table 19.2.) Note that not all the ^{40}K decays to ^{40}Ar.

* 56. Cobalt-60 and iodine-131 are used in treatments for some types of cancer. Cobalt-60 decays with a half-life of 5.27 years, emitting beta particles with a maximum energy of 0.32 MeV. Iodine-131 decays with a half-life of 8.04 days, emitting beta particles with a maximum energy of 0.60 MeV.
(a) Suppose a fixed small number of moles of each of these isotopes were to be ingested and remain in the body indefinitely. What is the *ratio* of the number of millisieverts of total lifetime radiation exposure that would be caused by the two radioisotopes?
(b) Now suppose that the contact with each of these isotopes is for a fixed short period, such as 1 hour. What is the ratio of millisieverts of radiation exposure for the two in this case?

57. Boron is used in control rods in nuclear power reactors because it is a good neutron absorber. When the isotope ^{10}B captures a neutron, an alpha particle (helium nucleus) is emitted. What other atom is formed? Write a balanced equation.

* 58. The average energy released in the fission of a ^{235}U nucleus is about 200 MeV. Suppose the conversion of this energy to electrical energy is 40% efficient. What mass of ^{235}U is converted to its fission products in a year's operation of a 1000-megawatt nuclear power station? Recall that 1 W is 1 J s^{-1}.

59. The energy released by a bomb is sometimes expressed in tons of TNT (trinitrotoluene). When one ton of TNT explodes, 4×10^9 J of energy is released. The fission of 1 mol of uranium releases approximately 2×10^{13} J of energy. Calculate the energy released by the fission of 1.2 kg of uranium in a small atomic bomb. Express your answer in tons of TNT.

60. The solar system abundances of the elements Li, Be, and B are four to seven orders of magnitude lower than those of the elements that immediately follow them: C, N, and O. Explain.

* 61. The sun's distance from earth is approximately 1.50×10^8 km, and the earth's radius is 6371 km. The earth receives radiant energy from hydrogen burning in the sun at a rate of 0.135 J s^{-1} cm^{-2}. Using the data of Table 19.1, calculate the mass of hydrogen converted per second in the sun.

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CUMULATIVE PROBLEMS

62. In 1951 wood from two sequoia trees was dated by the ^{14}C method. In one tree, clean borings located between the growth rings associated with the years A.D. 1057 and 1087 (that is, wood known to have grown 880 ± 15 years prior to the date of measurement) had a ^{14}C activity about 0.892 of that of wood growing in 1951. A sample from a second tree had an activity about 0.838 of that of new wood, and its age was established as 1377 ± 4 years by tree-ring counting.
    (a) What ages does carbon dating associate with the wood samples?
    (b) What values of $t_{1/2}$ can be deduced if the tree-ring dates given are used as the starting point?
    (c) Discuss assumptions underlying the calculations in (a) and (b), and indicate in what direction failures of these assumptions might affect the calculations.

63. A typical electrical generating plant has a capacity of 500 megawatts (MW; 1 MW = 10^6 J s^{-1}) and an overall efficiency of about 25%. (a) The combustion of 1 kg of bituminous coal releases about 3.2×10^4 kJ and leaves an ash residue of 100 g. What weight of coal must be used to operate a 500-MW generating plant for 1 year, and what weight of ash must be disposed of? (b) Enriched fuel for nuclear reactors contains about 4% ^{235}U, fission of which gives 1.9×10^{10} kJ per mole ^{235}U. What weight of ^{235}U is needed to operate a 500-MW power plant, assumed to have 25% efficiency, for 1 year, and what weight of fuel must be reprocessed to remove radioactive wastes? (c) The radiation from the sun striking the earth's surface on a sunny day corresponds to a power of 1.5 kW m^{-2}. How large must the collection surface be for a 500-MW solar-generating plant? (Assume that there are 6 hours of bright sun each day and that storage facilities continue to produce power at other times. The efficiency for solar-power generation would be about 25%.)

64. Examine the ratio of atomic mass to atomic number for the elements with *even* atomic number through calcium. This ratio is approximately the ratio of the average mass number to the atomic number.
    (a) Which two elements stand out as different in this set of ten?
    (b) What would be the "expected" atomic mass of argon, based on the correlation considered here?
    (c) Show how the anomaly in the ordering of natural atomic masses of argon and potassium can be accounted for by the formation of "extra" ^{40}Ar via decay of ^{40}K atoms.

65. Hydrazine, $N_2H_4(\ell)$, reacts with oxygen in a rocket engine to form nitrogen and water vapor:

$$N_2H_4(\ell) + O_2(g) \longrightarrow N_2(g) + 2\ H_2O(g)$$

    (a) Calculate $\Delta H°$ for this highly exothermic reaction at 25°C, using data from Appendix D.
    (b) Calculate $\Delta U°$ of this reaction at 25°C.
    (c) Calculate the total change in mass, in grams, during the reaction of 1.00 mol of hydrazine.

66. The long-lived isotope of radium, ^{226}Ra, decays by alpha particle emission to its daughter radon, ^{222}Rn, with a half-life of 1622 years. The energy of the alpha particle is 4.79 MeV. Suppose 1.00 g of ^{226}Ra, freed of all its radioactive progeny, were placed in a calorimeter that contained 10.0 g of water, initially at 25°C. Neglecting the heat capacity of the calorimeter and heat loss to the surroundings, calculate the temperature the water would reach after 1.00 hour. Take the specific heat of water to be 4.18 J K^{-1} g^{-1}.

67. The radioactive nuclide $^{232}_{90}Th$ has a half-life of 1.39×10^{10} years. It decays by a series of consecutive steps, the first two of which involve $^{228}_{88}Ra$ (half-life 6.7 years) and $^{228}_{89}Ac$ (half-life 6.13 hours).
    (a) Write balanced equations for the first two steps in the decay of ^{232}Th, indicating all decay products. Calculate the total kinetic energy carried off by the decay products.
    (b) After a short initial time, the rate of formation of ^{228}Ra becomes equal to its rate of decay. Express the number of ^{228}Ra nuclei in terms of the number of ^{232}Th nuclei, using the steady-state approximation from Section 18.4.

68. Zirconium is used in the fuel rods of most nuclear power plants. The following half-cell reduction potential applies to aqueous acidic solution:

$$ZrO_2(s) + 4\ H_3O^+(aq) + 4e^- \longrightarrow Zr(s) + 6H_2O(\ell) \qquad E° = -1.43\ V$$

    (a) Predict whether zirconium can reduce water to hydrogen. Write a balanced equation for the overall reaction.
    (b) Calculate $E°$ and K for the reaction in part (a).
    (c) Can your answer to part (b) explain the release of hydrogen in the Three Mile Island accident and the much greater release of hydrogen (which subsequently exploded) at Chernobyl?

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